MATH256
Linear Algebra HW#2

1. (1.4.14) Let $\mathbf{u} = \begin{bmatrix} 4 \\ -1 \\ 4 \end{bmatrix}$ and $A = \begin{bmatrix} 2 & 5 & -1 \\ 0 & 1 & -1 \\ 1 & 2 & 0 \end{bmatrix}$. Is $\mathbf{u}$ in the subset of $\mathbb{R}^3$ spanned by the columns of A ? Why or why not ?

2. (1.4.16) Let $A = \begin{bmatrix} 1 & -2 & -1 \\ -2 & 2 & 0 \\ 4 & -1 & 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$. Show that the equation $\mathbf{Ax} = \mathbf{b}$ does not have a solution for all possible $\mathbf{b}$, and describe the set of all $\mathbf{b}$ for which $\mathbf{Ax} = \mathbf{b}$ does have a solution

3. (1.4.23) Mark each statement True or False. Justify each answer.

a. The equation $\mathbf{Ax} = \mathbf{b}$ is referred to as a *vector equation* F matrix equation
b. A vector b is a linear combination of the columns of a matrix. A if an only if the equation $\mathbf{Ax} = \mathbf{b}$ has at least one solution. T
c. The equation $\mathbf{Ax} = \mathbf{b}$ is consistent if the argumented matrix $[A \quad \boldsymbol{b}]$ has a pivot position in every row T
d. The first entry in the product $\mathbf{Ax}$ is a sum of products.
e. If the columns of an *m x n* matrix A span $\mathbb{R}^m$, then the equation $\mathbf{Ax} = \mathbf{b}$ is consistent for each $\mathbf{b}$ in $\mathbb{R}^m$. T
f. If A is an *m x n* matrix and if the equation $\mathbf{Ax} = \mathbf{b}$ is inconsistent for some b in $\mathbb{R}^m$,then A cannot have a pivot position in every row. T

4. (1.4.24) Mark each statement True or False .Justify each answer.

a. Every matrix equation $\mathbf{Ax} = \mathbf{b}$ corresponds to a vector equation with the same solution set . T
b. If the equation $\mathbf{Ax} = \mathbf{b}$ is consistent , then $\mathbf{b}$ is in the set spanned by the columns of A. T
c. Any linear combination of vectors can always be written in the form Ax for a suitable matrix A and vector x. T
d. If the coefficient marix A has a pivot position in every row , then the equation $\mathbf{Ax} = \mathbf{b}$ is inconsistent F CONSISTENT
e. The solution set of a linear system whose augmented matrix is $[\boldsymbol{a_1} \quad \boldsymbol{a_2} \quad \boldsymbol{a_3} \quad \boldsymbol{b}]$ is the same as the solution set of $\mathbf{Ax} = \mathbf{b}$, if $A = [\boldsymbol{a_1} \quad \boldsymbol{a_2} \quad \boldsymbol{a_3}]$. T
f. If A is an *m x n* matrix whose columns do not span $\mathbb{R}^m$, then the equation $\mathbf{Ax} = \mathbf{b}$ is consistent for every $\mathbf{b}$ in $\mathbb{R}^m$. F maybe INCONSISTENT

5. (1.5.2) Determine if the system has a nontrivial solution. Try to use as few row operations as possible .

$$\begin{aligned} x_1 - 2x_2 + 3x_3 &= 0 \\ -2x_1 - 3x_2 - 4x_3 &= 0 \\ 2x_1 - 4x_2 + 9x_3 &= 0 \end{aligned}$$

6. (1.5.4) Determine if the system has a nontrivial solution. Try to use as few row operations as possible .

$$\begin{aligned} 5x_1 - 3x_2 + 2x_3 &= 0 \\ -3x_1 - 4x_2 + 2x_3 &= 0 \end{aligned}$$

7. (1.7.6) Determine if the columns of the matrix form a linearly independent set. Justify your answer.

$$\begin{bmatrix} -4 & -3 & 0 \\ 0 & -1 & 5 \\ 1 & 1 & -5 \\ 2 & 1 & -10 \end{bmatrix}$$

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8. (1.7.8) Determine if the columns of the matrix form a linearly independent set. Justify your answer.

$$\begin{bmatrix} 1 & -2 & 3 & 2 \\ -2 & 4 & -6 & 2 \\ 0 & 1 & -1 & 3 \end{bmatrix}$$

9. (1.7.10) $\boldsymbol{v_1} = \begin{bmatrix} 1 \\ -3 \\ -5 \end{bmatrix}$, $\boldsymbol{v_2} = \begin{bmatrix} -3 \\ 9 \\ 15 \end{bmatrix}$, $\boldsymbol{v_3} = \begin{bmatrix} 2 \\ -5 \\ h \end{bmatrix}$

a) For what values of h is $\boldsymbol{v_3}$ in Span $\{\boldsymbol{v_1}, \boldsymbol{v_2}\}$, and
b) For what values of h is $\{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}\}$ linearly dependent? Justify each answer.